Linux File System - Detailed Explanation

Linux uses a hierarchical file system that is different from Windows. Here's a detailed breakdown:

1 Root Directory /

- The top-most directory in Linux is /, called the root directory.
- All files and folders in Linux exist under /. There are no drive letters like Windows.
- Think of / as the "trunk of a tree" with all directories branching out from it.

2 Key System Directories

Directory	Purpose	Example/Notes
/home	User home directories	/home/vardhan → personal files, configs, folders
/root	Home directory for root (admin)	Only accessible by root user
/bin	Essential binary executables	Commands like ls, cp, mv
/sbin	System binaries for admin tasks	Commands like fdisk, ifconfig
/usr	User programs and utilities	/usr/bin → installed software commands, /usr/lib → libraries
/etc	System configuration files	/etc/passwd, /etc/ssh/sshd_config
/var	Variable files	Logs (/var/log), caches, mail, databases
/tmp	Temporary files	Cleared on reboot
/dev	Device files	Represents hardware like disks, USB drives
/mnt	Mount points for external drives	Windows drives appear here in WSL: /mnt/c, /mnt/d
/opt	Optional software	Third-party software installations
/lib	Essential libraries	Needed by commands in /bin and /sbin
/media	Removable media	USB drives, CDs mounted here

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3 File Paths

- **Absolute Path:** Starts from root /. Example: /home/vardhan/Documents/file.txt

- **Relative Path:** Based on current directory. Example: Documents/file.txt if in /home/vardhan
- **Special Shortcuts:**
  - . → Current directory
  - .. → Parent directory
  - ~ → Your home directory (/home/vardhan)

4️⃣ Case Sensitivity

- Linux filenames are case-sensitive: Desktop ≠ desktop ≠ DESKTOP

5️⃣ Permissions

- Linux uses **read (r), write (w), execute (x)** permissions
- **User categories:** owner, group, others
- Example from ls -l:

-rwxr--r-- 1 vardhan vardhan 1024 Oct 13 20:00 myfile

- Owner (vardhan) can read, write, execute
- Group can read
- Others can read

6️⃣ Mounting Drives

- Drives are mounted as directories, not letters.
- Example in WSL: C: → /mnt/c, D: → /mnt/d
- Navigate like:

cd /mnt/c/Users/DEVA\ SATYA\ VARDHAN/Desktop

7️⃣ Hidden Files

- Files starting with . are hidden, usually configs: .bashrc, .profile
- List them with:

ls -a

8️⃣ File Types

- Regular files → -

- Directory → d
- Link → l
- Device → c or b
- Example:

drwxr-xr-x 2 vardhan vardhan 4096 Oct 13 20:00 Documents

-rw-r--r-- 1 vardhan vardhan 1024 Oct 13 20:00 file.txt

Summary

- Linux uses a **single-root hierarchical structure**.
- Everything is a **file or directory**, including devices.
- **Permissions and case sensitivity** are strictly enforced.
- Drives and external devices are **mounted as directories**.
- WSL allows accessing Windows drives under /mnt.

This structure allows Linux to be flexible, secure, and powerful for both users and system administration.